

Largest Semiconductor  
Products Manufacturer  
↳ Nvidia

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Largest Electronics

Products Company  
↳ Apple (By Market Cap)

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Jensen Huang

↳ CEO (Nvidia)

Lisa Su

↳ CEO (AMD)

Morris Chang

↳ Founder (TSMC)

Cristiano Amon

↳ CEO (Qualcomm)

Sanjay Mehrotra

↳ CEO (Micron)

IC Bose,

credited in

Shockley's  
speech for

making the first transistor

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IDM v/s Fab v/s Design

# Correlation b/w Electronics & Semicon industry in India

|         |                |      |                 |
|---------|----------------|------|-----------------|
| Elec    | 200B           | 500B | 3T              |
| Semicon | 40B            | 135B | 810B            |
|         | 2023           | 2030 | 2047            |
|         | 5.8%<br>of GDP |      | 10.2%<br>of GDP |

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Products & Technology are essential for properly growing economy & strategic autonomy.

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90% consumption from Global brands from electronics PoV

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→ Technology sovereignty  
(can be shut-off anytime)

→ Economic Self-reliance

(Allows us to  
reinvest our country's  
consumption in our  
R&D)

→ Components ecosystem

(If we do products,  
our manufacturers  
will also have a larger  
viable market.)

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Electronics Products Growth

Domains

1) AI/ML \*\*\*

2) Auto/EV

3) Consumer Electronics

4) Telecom & Wireless

5) Medical Bio-Tech

6) Energy (Clean Tech)

7) Space Defense

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AI & Semiconductor

inter-connection

| 1980's                                                         | Now                                                            |
|----------------------------------------------------------------|----------------------------------------------------------------|
| Required supercomputers to simulate as small as 500-node NN's. | Pocket-sized devices managing LLM's w/ trillions of parameters |

# → Emerging Trends "7"

- Security
- Custom Chips (Google / Microsoft / NVIDIA / Amazon)
- Architecture Explosion
- Advanced 3D Packaging & Chiplets

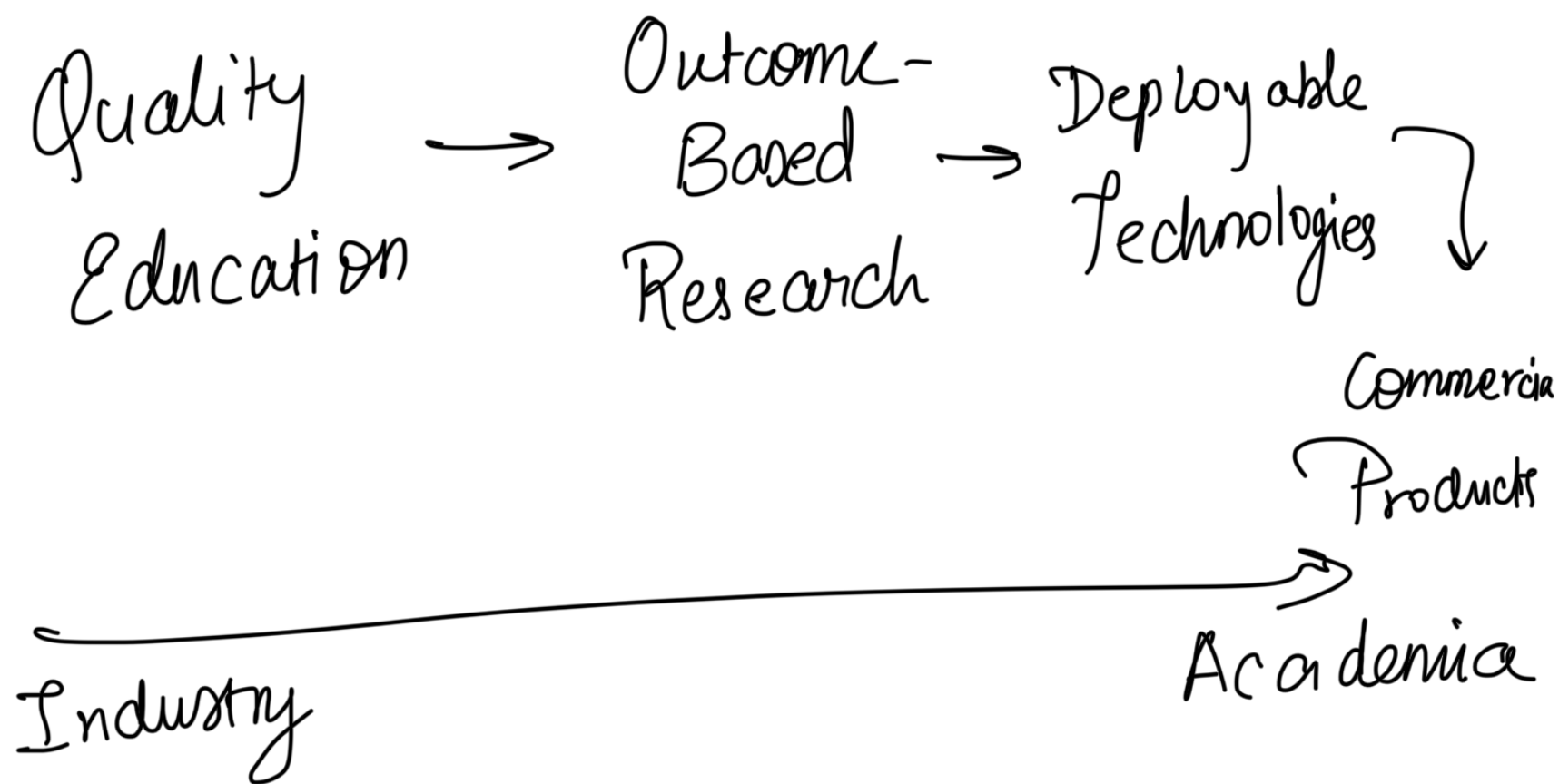
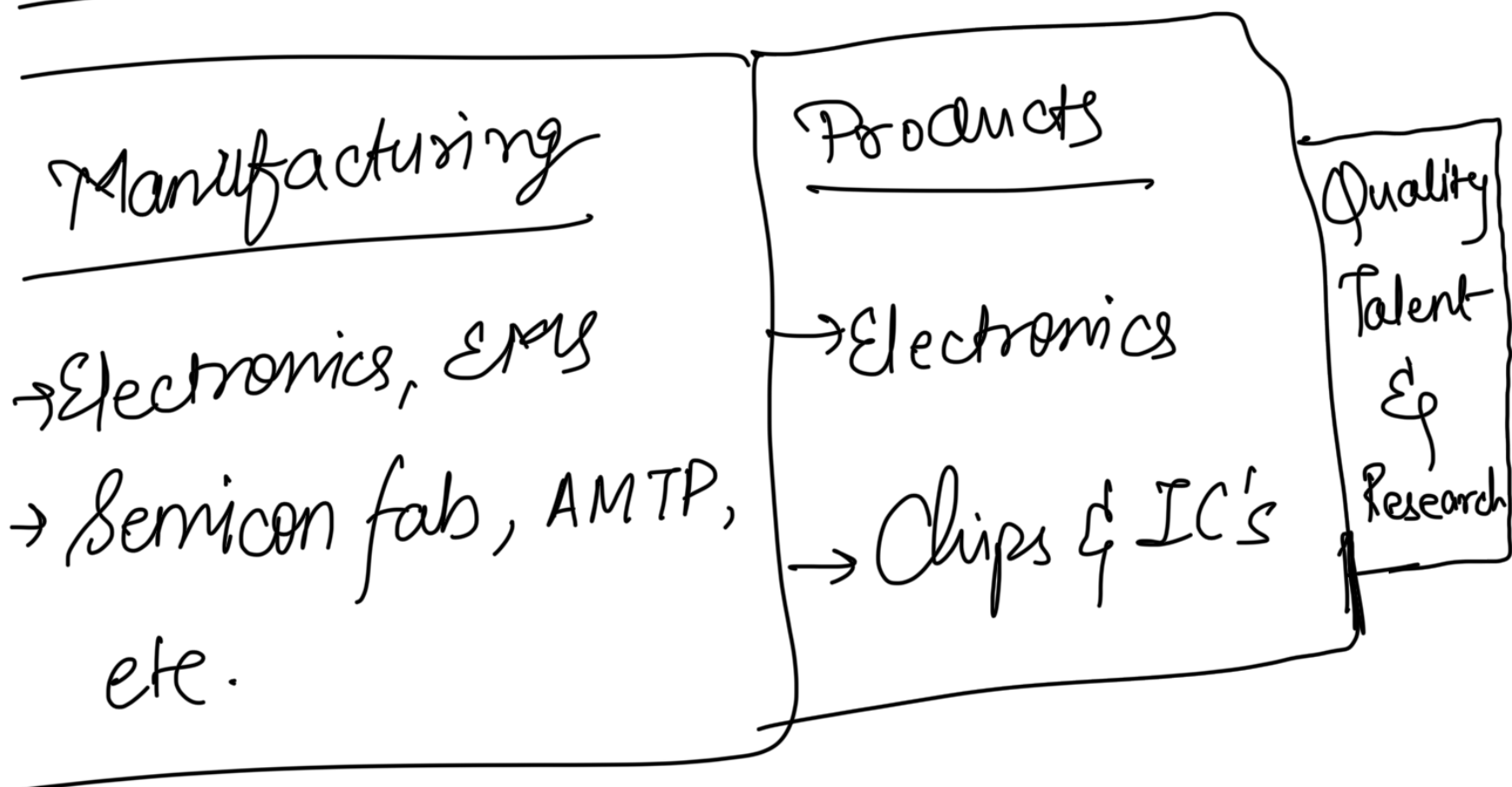
↳ We need vertical integration

Photomask & Reticles

↳ 2e arm work & dimensions

AI causing them to reach limit

- Wireless functionality
  - Photonics based interconnect b/w chips
  - Heterogeneous & vertical integration
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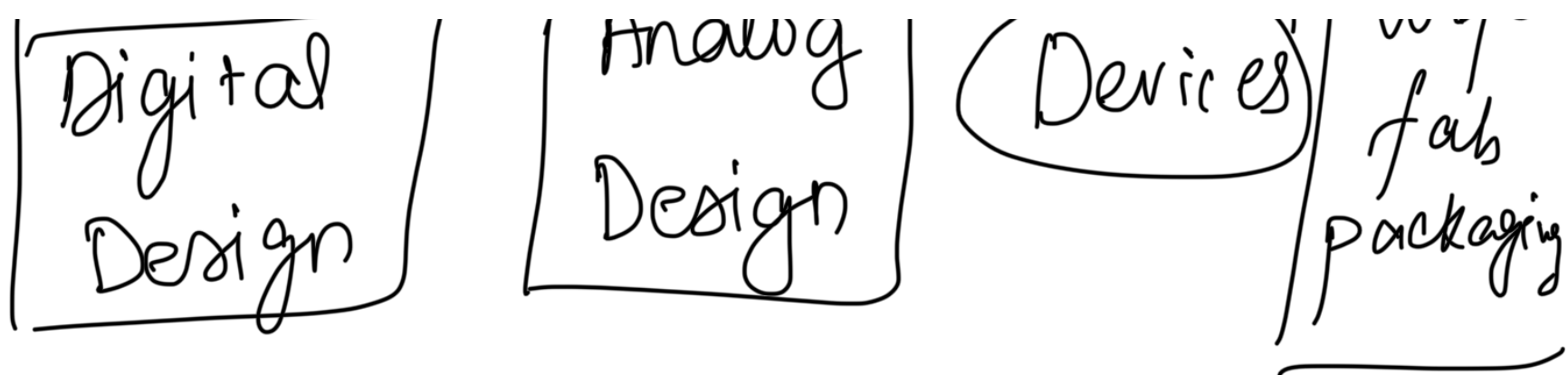
## Semiconductor Industry

Semiconductor Chips

Manufacturing







NVIDIA, Qualcomm, etc.

SMEC, IBM,  
TSMC, Micron,  
etc.

- RTL Design
- Verification
- Physical & Digital Design

Build fundamental understanding using complicated projects rather than remaining pedagogy focused

|                           |                   |
|---------------------------|-------------------|
| <del>32-bit address</del> | Make a whole chip |
|---------------------------|-------------------|

X-86 (Intel), ARM  $\rightarrow$  Proprietary  
Architecture

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RISC-V  $\rightarrow$  Open-source,  
customisable  
architecture

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